| Semester: V | | | | | | |
|---|---|---|---|---|---|---|
| **ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING**<br>**Category: PROFESSIONAL CORE COURSE**<br>**(Theory and Lab)**<br>**(Common to CS, CD,CY & IS)** | | | | | | |
| **Course Code** | **:** | **IS353IA** | | **CIE** | **:** | **100 + 50 Marks** |
| **Credits: L:T:P** | **:** | **3:0:1** | | **SEE** | **:** | **100 + 50 Marks** |
| **Total Hours** | **:** | **45L + 30P** | | **SEE Duration** | **:** | **3 + 3 Hours** |

| **Unit-I** | **09 Hrs** |
|---|---|
| **Introduction:** What is AI?<br>Intelligent agents: Intelligent Agents: Agents and environment; Rationality; the nature of environments; the structure of agents<br>**Problem Solving & Uninformed Search Strategies:** Problem-solving agents, Breadth-first Search, Depth-first Search, Depth-limited Search and Iterative Deepening Depth First Search. | |
| **Unit – II** | **09 Hrs** |
| **Informed (Heuristic) Search Strategies:** A*Search, Heuristic Functions<br>**Beyond Classical Search:** Local Search Algorithms and Optimization Problems, Hill-climbing Search, Simulated Annealing, Local-beam Search, Genetic Algorithms<br>**Adversarial search:** Games, Optimal decision in games, Alpha-Beta Pruning | |
| **Unit –III** | **09 Hrs** |
| **Supervised Learning:** Basic Concepts, General Framework for Classification<br>**Decision Tree Classifier**-A Basic Algorithm to Build a Decision Tree, Methods for Expressing Attribute Test Conditions, Measures for Selecting an Attribute Test Condition, Algorithm for Decision Tree Induction, Characteristics of Decision Tree Classifiers,<br>**Model Overfitting**- Reasons for Model Overfitting<br>**Model Selection** - Using a Validation Set, Incorporating Model Complexity, Estimating Statistical Bounds, Model Selection for Decision Trees, Model Evaluation | |
| **Unit –IV** | **09 Hrs** |
| **Nearest Neighbor Classifiers**-Characteristics of Nearest Neighbor Classifiers<br>**Naive Bayes Classifier**-Basics of Probability Theory, Naive Bayes assumption<br>**Logistic Regression**-Logistic Regression as a Generalized Linear Model, Learning Model Parameters, Characteristics of Logistic Regression<br>**Ensemble Methods** – Methods for constructing Ensemble classifier, Bagging, Boosting, Random Forests | |
| **Unit –V** | **09 Hrs** |
| **Unsupervised Learning**- Overview, What Is Cluster Analysis, Different Types of Clustering's, Different Types of Clusters<br>**K-means**-The Basic K-means Algorithm, Additional Issues, Bisecting K-means, K-means and Different Types of Clusters, Strengths and Weaknesses, K-means as an Optimization Problem<br>**Cluster Evaluation**-Overview, Unsupervised Cluster Evaluation Using Cohesion and Separation, Unsupervised Cluster Evaluation Using the Proximity Matrix, Determining the Correct Number of Clusters, Supervised Measures of Cluster Validity, Assessing the Significance of Cluster Validity Measures, Choosing a Cluster Validity Measure | |

| Semester: V | | | | | | |
|---|---|---|---|---|---|---|
| **DATABASE MANAGEMENT SYSTEMS**<br>**Category: PROFESSIONAL CORE COURSE**<br>**(Theory and Lab)**<br>(Common to CS & IS, AI, CD) | | | | | | |
| **Course Code** | **:** | **CD252IA** | | **CIE** | **:** | **100 + 50 Marks** |
| **Credits: L:T:P** | **:** | **3:0:1** | | **SEE** | **:** | **100 + 50 Marks** |
| **Total Hours** | **:** | **45L+30P** | | **SEE Duration** | **:** | **3 + 3 Hours** |

| **Unit-I** | **09 Hrs** |
|---|---|
| **Introduction to Database Systems -**Databases and Database users: Introduction, An example, Characteristics of Database Approach,Data Models, Schemas and Instances, Three-schema Architecture and Data Independence, The Database System Environment.<br><br>**Data Modeling Using the Entity-Relationship Model-** High-Level Conceptual Data Models for Database Design; A Sample Database Application; Entity Types, Entity Sets, Attributes and Keys; Relationship types, Relationship Sets, Roles and Structural Constraints; Weak Entity Types. | |
| **Unit – II** | **09 Hrs** |
| **Refining the ER Design for the COMPANY Database;** ER Diagrams, Naming Conventions and Design Issues, ER- to-Relational Mapping.<br>**Relational Model and Relational Algebra-**Relational Model Concepts; Relational Model Constraints and Relational Database Schemas; Update Operations and Dealing with Constraint Violations; Unary Relational Operations: SELECT and PROJECT; Relational Algebra Operations from Set Theory; Binary Relational Operations: JOIN and DIVISION ; Examples of Queries in Relational Algebra. | |
| **Unit –III** | **09 Hrs** |
| **Introduction to SQL- S**QL Data Definition, Specifying Constraints in SQL, Basic Queries in SQL; Insert, Delete and Update Statements in SQL More Complex SQL Retrieval Queries.<br>**Relational Database Design -** Functional Dependencies – Definition, Inference Rules, Equivalence of sets of FD's, Minimal Set of FD's ; Normal Forms Based on Primary Keys; General Definitions of Second and Third Normal Forms; Boyce-Codd Normal Form;Properties of Relational Decompositions. | |
| **Unit –IV** | **09 Hrs** |
| **Transaction Processing Concepts-** Introduction to transaction processing, Transaction states and additional operations, Desirable properties of transaction, Schedules of transactions. Characterizing schedules based on Serializability: Serial, Non serial and Conflict- Serializable schedules, Testing for Conflict serializability of schedule<br>**Concurrency Control Techniques:** Two phase locking techniques for concurrency control, types of locks and system lock tables | |
| **Unit –V** | **09 Hrs** |
| **Introduction to NoSQL**: Aggregate data models: aggregates, key-value and document data models. Distribution models: sharding , master-slave replication, peer-peer replication – combining sharding and replication.<br>Big Data : Types of data: Structured, semi structured, unstructured. Distributed Architectures : Hadoop, Map Reduce Programming Model | |

| Semester: V | | | | | | |
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| **THEORY OF COMPUTATION**<br>**Category: PROFESSIONAL CORE COURSE**<br>**(Theory)**<br>**(Common to CS, CD, CY & IS)** | | | | | | |
| **Course Code** | **:** | **CS354TA** | | **CIE** | **:** | **100 Marks** |
| **Credits: L:T:P** | **:** | **3:1:0** | | **SEE** | **:** | **100 Marks** |
| **Total Hours** | **:** | **45L + 30T** | | **SEE Duration** | **:** | **3 Hours** |

| **Unit-I** | **09 Hrs** |
|---|---|
| Regular Languages and Regular Expressions, Memory Required to Recognize a Language, Deterministic Finite Automata (DFA), Non Deterministic Finite Automata (NFA), Non Deterministic Finite Automata with ϵ -transitions (NFA-ϵ), Equivalence, Regular Expressions and Finite Automata, Applications of Regular Expressions, Algebraic laws of Regular Expressions, Minimization of Finite Automata. | |
| **Unit – II** | **09 Hrs** |
| Pumping Lemma for Regular Languages, Closure properties of Regular Languages, Decision properties of Regular languages. Context-free grammars (CFG), Parse trees, Applications, Ambiguity in grammars & languages, Simplification of CFG, Normal forms of CFGs. Regular Grammars, Equivalence of Regular Grammars and Finite Automata. | |
| **Unit –III** | **09 Hrs** |
| Push Down Automata (PDA): Definition, the languages of a PDA, Equivalence of PDA's & CFG's, Deterministic PDA. The Pumping Lemma for Context Free Languages (CFL), Closure properties of CFLs, Decision properties of CFLs | |
| **Unit –IV** | **09 Hrs** |
| Context Sensitive Languages (CSL) and Linear Bounded Automata (LBA), Turing Machines (TM): Definitions and Examples, TM as a Language Accepter, Computing Partial Functions with Turing Machine, Variations of Turing Machines, Combining Turing Machines, Non Deterministic TM, Universal TM. | |
| **Unit –V** | **09 Hrs** |
| Recursively Enumerable Languages (REL) and Recursive Languages. Properties of REL and Recursive Languages. More General Grammars: Context Sensitive Grammar and Unrestricted Grammar, Chomsky Hierarchy, Not all languages are Recursively Enumerable, Unsolvable Problem, Reducing One problem to another, The halting problem of TM, Post's Correspondence Problem (PCP), Time and Space Complexity of TM. | |

| Semester : V | | | | | | |
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| **PRINCIPLES OF MANAGEMENT & ECONOMICS (Theory)** | | | | | | |
| **Course Code** | **:** | HS251TA | | **CIE** | **:** | **100 Marks** |
| **Credits: L:T:P** | **:** | 3:0:0 | | **SEE** | **:** | **100 Marks** |
| **Total Hours** | **:** | 45 Hrs | | **SEE Duration** | **:** | **3.00 Hours** |

| **Unit-I** | **06 Hrs** |
|---|---|
| **Introduction to Management:** Management Functions – POSDCORB – an overview, Management levels & Skills, Management History - **Classical Approach:** Scientific Management, Administrative Theory, **Quantitative Approach:** Operations Research, **Behavioral Approach:** Hawthorne Studies, **Contemporary Approach:** Systems Theory, Contingency Theory. **Caselets / Case studies** | |
| **Unit – II** | **10 Hrs** |
| **Foundations of Planning:** Types of Goals & Plans, Approaches to Setting Goals & Plans, Strategic Management Process, Corporate strategies – types of corporate strategies, BCG matrix, Competitive Strategies – Porters Five force Model, types of Competitive Strategies. **Caselets / Case studies**<br>**Organizational Structure & Design:** Overview of Designing Organizational Structure - Work Specialization, Departmentalization, Chain of Command, Span of Control, Centralization & Decentralization, Formalization, Mechanistic & Organic Structures. **Caselets / Case studies** | |
| **Unit –III** | **10 Hrs** |
| **Motivation:** Early Theories of Motivation - Maslow's Hierarchy of Needs Theory, McGregor's Theory X & Theory Y, Herzberg's Two Factor Theory. Contemporary Theories of Motivation: Adam's Equitytheory, Vroom's Expectancy Theory. **Caselets / Case studies**<br>**Leadership:** Behavioral Theories: Blake & Mouton's Managerial Grid, Contingency Theories of Leadership: Hersey & Blanchard's Situational Leadership, Contemporary Views of Leadership: Transactional & Transformational Leadership. **Caselets / Case studies** | |
| **Unit –IV** | **10 Hrs** |
| **Introduction to Economics:** Microeconomics and Macroeconomics, Circular flow model of economics, An Overview of Economic Systems.<br>**Essentials of Microeconomics:** Demand, Supply, and Equilibrium in Markets for Goods and Services, Price Elasticity of Demand and Price Elasticity of Supply, Elasticity and Pricing, Numericals on determining price elasticity of demand and supply. Changes in Income and Prices Affecting Consumption Choices, Monopolistic Competition, Oligopoly. | |
| **Unit –V** | **09 Hrs** |
| **Macroeconomic Indicators:** Prices and inflation, Consumer Price Index, Exchange rate, Labor Market, Money and banks, Interest rate. Gross Domestic product (GDP) - components of GDP, Measures of GDP: Outcome Method, Income method and Expenditure method, Numericals on GDP Calculations, ESG an overview.<br>**Macroeconomic models**- The classical growth theory, Keynesian cross model, IS-LM-model, The AS-AD model, The complete Keynesian model, The neo-classical synthesis. National Budgeting process in India | |

| Semester: V | | | | | | |
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| **Vulnerability Assessment & Penetration Testing**<br>**Category: PROFESSIONAL CORE COURSE ELECTIVE-I**<br>**(Group-B)**<br>**(Theory and Lab)** | | | | | | |
| **Course Code** | **:** | **CY255TBD** | | **CIE** | **:** | **100 Marks** |
| **Credits: L:T:P** | **:** | **3:0:0** | | **SEE** | **:** | **100 Marks** |
| **Total Hours** | **:** | **45L** | | **SEE Duration** | **:** | **3 Hours** |

| **Unit-I** | **09 Hrs** |
|---|---|
| **Introduction to Vulnerability Assessment & Penetration Testing:** Why you need to understand your enemy's tactics, recognizing the gray areas in security, Vulnerability Assessment and Penetration Testing. Penetration Testing and Tools: Social Engineering Attacks: How a social engineering attack works, conducting a social engineering attack, common attacks used in penetration testing, preparing yourself for face-to-face attacks, defending against social engineering attacks. | |
| **Unit – II** | **09 Hrs** |
| **Physical Penetration Attacks:** Why a physical penetration is important, conducting a physical penetration, Common ways into a building, Defending against physical penetrations. Insider Attacks: Conducting an insider attack, Defending against insider attacks. Metasploit: The Big Picture, Getting Metasploit, Using the Metasploit Console to Launch Exploits, Exploiting Clients Side Vulnerabilities with Metasploit, Penetration Testing with Metasploit's Meterpreter, Automating and Scripting Metasploit, Going Further with Metasploit | |
| **Unit –III** | **09 Hrs** |
| **Managing a Penetration Test:** planning a penetration test, structuring a penetration test, execution of a penetration test, information sharing during a penetration test, reporting the results of a Penetration Test. Basic Linux Exploits: Stack Operations, Buffer Overflows, Local Buffer Overflow Exploits, Exploit Development Process. Windows Exploits: Compiling and Debugging Windows Programs, Writing Windows Exploits, Understanding Structured Exception Handling (SEH), Understanding Windows Memory Protections (XPSP3, Vista 7 and Server2008), By passing Windows Memory Protections. | |
| **Unit –IV** | **09 Hrs** |
| **Web Application Security Vulnerabilities:** Overview of top web application security vulnerabilities, Injection vulnerabilities, cross-Site scripting vulnerabilities, the rest of the OWASP Top Ten SQL Injection vulnerabilities, Cross-site scripting vulnerabilities. Vulnerability Analysis: Passive Analysis, Source Code Analysis, Binary Analysis. | |
| **Unit –V** | **09 Hrs** |
| **Client-Side Browser Exploits:** Why client-side vulnerabilities are interesting, Internet explorer security concepts, history of client- side exploits and latest trends, finding new browser-based vulnerabilities heap spray to exploit, protecting your self from clients side exploit. Malware Analysis: Collecting Malware and Initial Analysis: Malware, Latest Trends in Honeynet Technology, Catching Malware: Setting the Trap, Initial Analysis of Malware. | |